

# Intelligent Document Extraction, Validation & API Platform

Invoices · Balance Sheets · Profit & Loss · Cash Flow Statements

OCR + layout-aware extraction · financial reconciliation · REST API · dashboard

# The problem

The same numbers arrive in inconsistent, often scanned, layouts

1

## Validate

Type, integrity and page limit — before OCR

2

## Read

Native PDF text, or rasterise and OCR

3

## Extract

Fields and every table row, grounded

4

## Reconcile

Recompute the financial relationships

5

## Serve

Store, expose via API and dashboard

### Four document types in scope

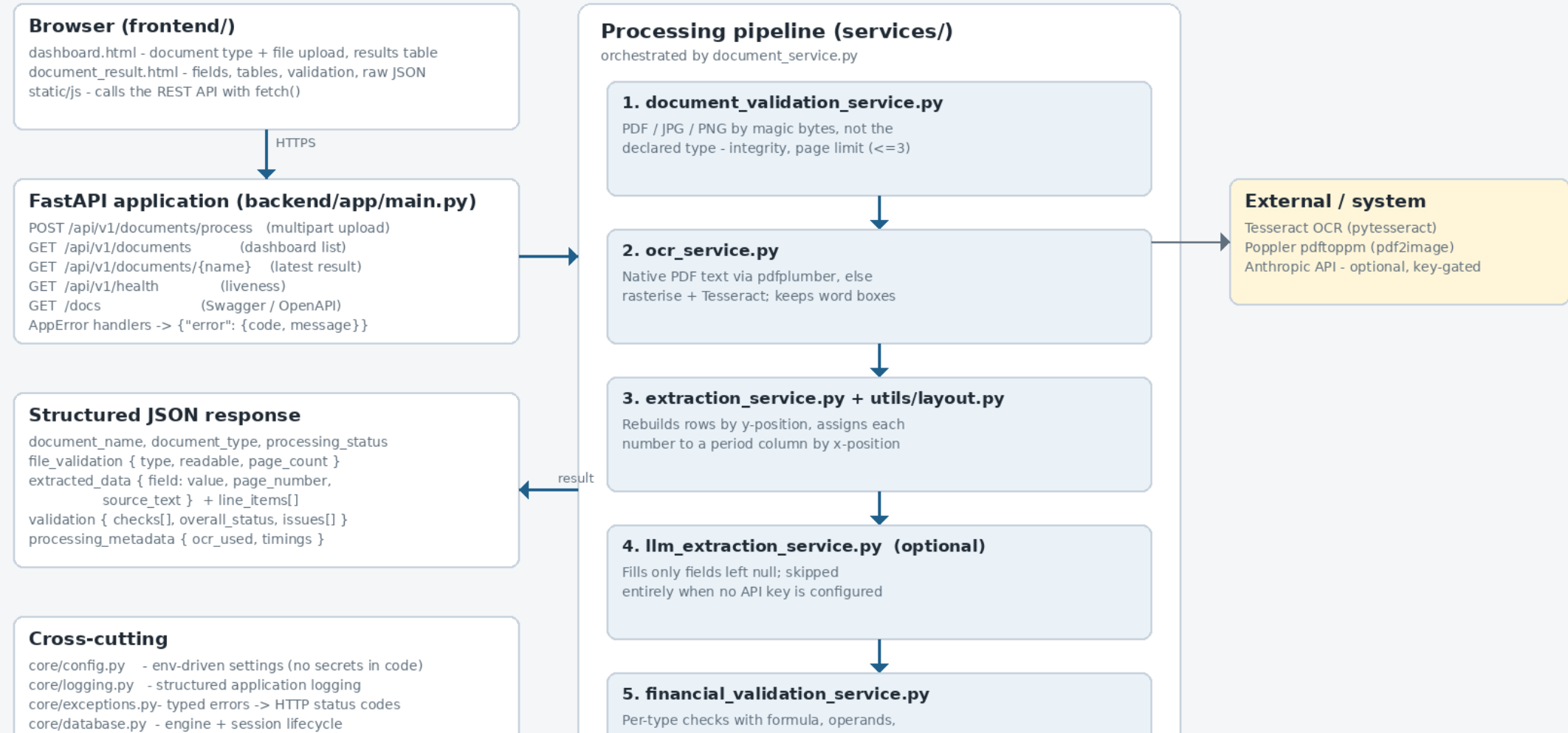
- Invoice — header fields, line items, tax, cash and change
- Balance sheet — capital & liabilities and assets, per period
- Profit & loss — income, expenditure, profit and appropriations
- Cash flow — operating, investing, financing, opening/closing cash

# Architecture

One FastAPI application serves the API and the dashboard

## Intelligent Document Extraction, Validation & API Platform

Request path for POST /api/v1/documents/process



# Why line-by-line OCR text is not enough

The insight that drove the design

## Reading OCR text as lines

Total      743, 732, 155

A row filled in for only one year gives no clue which year it belongs to — and OCR often emits labels and numbers as separate blocks entirely.

## Reading the page as a table

Total · x≈1540 → column 2 → 31-Mar-16

Word boxes are kept from both PDF text and OCR. Rows are rebuilt by vertical position; each number is assigned to a period column by its right edge.

## Measured, not assumed

- Benchmarked all 30 dataset statements; the first extractor was over-fitted to one year
- Compared page-segmentation modes and DPI — 300 DPI measured worse than 200
- Extraction and validation had drifted to two different sets of field lookups; unified them
- A second colour OCR pass gained one check for twice the latency, so it was left out

# Financial validation

Each check reports formula, operands, calculated vs reported, variance, status

## Invoice

$qty \times unit\ price \approx line\ amount \cdot \sum lines \approx subtotal \cdot subtotal + tax - discount \approx total \cdot cash - total \approx change$

## Balance sheet

$Total\ Capital\ \&\ Liabilities \approx Total\ Assets \cdot \sum component\ rows \approx each\ reported\ total$

## Profit & loss

$Interest + Other\ Income \approx Total\ Income \cdot Income - Expenditure \approx Net\ Profit \cdot - Minority \approx Attributable$

## Cash flow

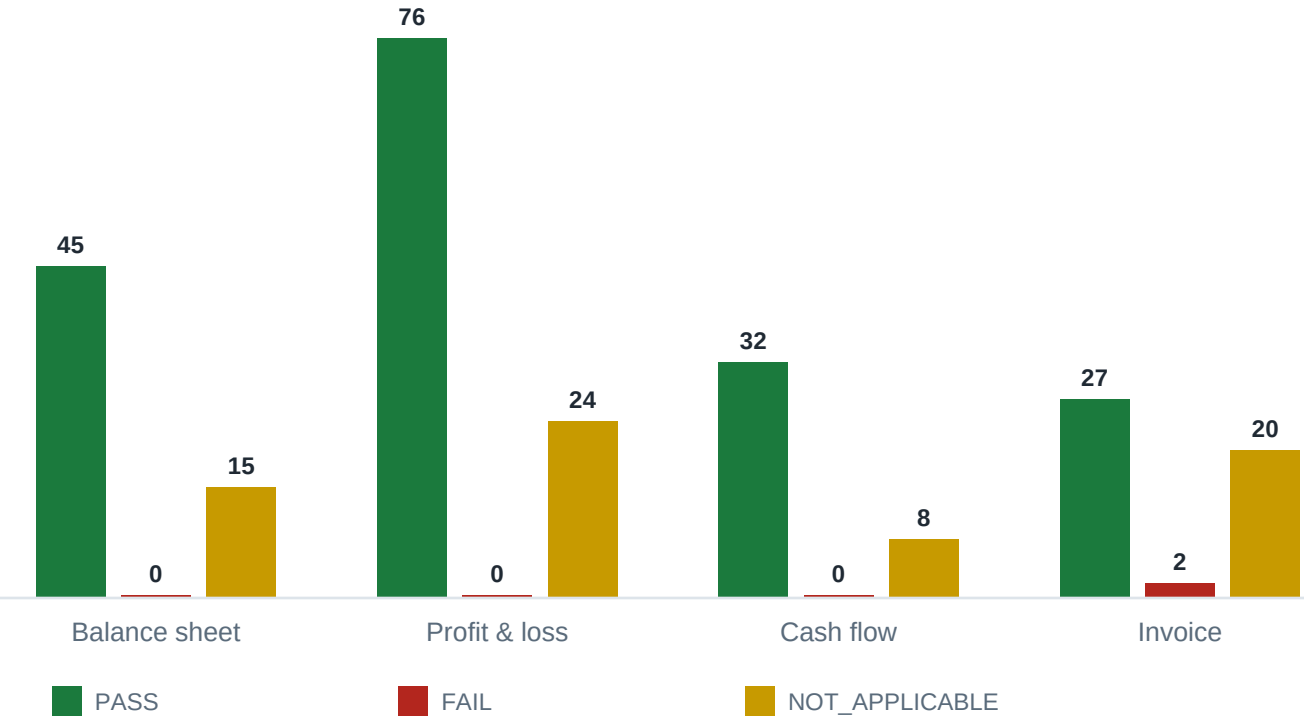
$Operating + Investing + Financing + FX \approx Net\ Increase \cdot Opening + Net\ Increase \approx Closing\ Cash$

## Honest by design

- Missing operand → NOT\_APPLICABLE, never a failure
- A partly-read section is not summed — an unread row would count as zero
- Tolerance: max(1.0, 1% of the larger value)
- Bracketed figures parse as negative
- Every check runs once per reporting period
- Missing values stay null — never guessed

# Measured results

All 50 documents in the provided dataset, 297 checks



228

checks reconcile exactly

2

genuine discrepancies, verified by hand

67

not applicable — nothing to reconcile

Clean scans reconcile fully; the not-applicable results concentrate in the 2020–2022 scans where OCR cannot recover row labels.

# API and dashboard

One deployment serves both; Swagger is generated from the code

|             |                           |                    |
|-------------|---------------------------|--------------------|
| <b>POST</b> | /api/v1/documents/process | upload and process |
| <b>GET</b>  | /api/v1/documents         | dashboard list     |
| <b>GET</b>  | /api/v1/documents/{name}  | latest result      |
| <b>GET</b>  | /api/v1/health            | health check       |
| <b>GET</b>  | /docs                     | Swagger / OpenAPI  |

## Errors return one envelope

```
{ "error": { "code": "UNSUPPORTED_FILE_TYPE",
             "message": "Only PDF / JPG / PNG..." } }
```

415 · 413 · 400 · 404 · 422 · 500 — never a stack trace

Intelligent Document Extraction & Validation API

Back to dashboard

X51005361895.jpg

Type: invoice · Status: PASS · Processed at: 9/10/2026, 5:10:40 PM · OCR used: yes

Extracted fields

VENDOR NAME

Not found

VENDOR ADDRESS

Not found

VENDOR TAX ID

Not found

CUSTOMER NAME

Not found

INVOICE NUMBER

1CRO576494 (p.1)

INVOICE DATE

25/01/2018 (p.1)

INVOICE TIME

Not found

CURRENCY

Not found

SUBTOTAL

Not found

TAX AMOUNT

0.51 (p.1)

TAX RATE PERCENT

6 (p.1)

TAX INCLUSIVE

true

DISCOUNT

Not found

TOTAL AMOUNT

9 (p.1)

TOTAL QUANTITY

Not found

CASH PAID

50 (p.1)

CHANGE

41 (p.1)

Financial validation

Overall status: PASS

| Check                     | Formula                                                                                  | Calculated | Reported | Variance | Status            |
|---------------------------|------------------------------------------------------------------------------------------|------------|----------|----------|-------------------|
| tax_inclusive_total_check | (total_amount - tax_amount) + tax_amount ≈ total_amount (tax shown as included in total) | 9          | 9        | 0        | <span>PASS</span> |
| cash_change_check         | cash_paid - total_amount ≈ change                                                        | 41         | 41       | 0        | <span>PASS</span> |

Raw JSON

Show / hide

# Limitations and what I'd change

## Known today

- OCR quality is the binding constraint — the 2020–2022 scans lose row labels entirely
- Statement parsing is tuned to this dataset's banking format
- Noisy receipts yield fewer line items
- Synchronous processing, ~2–6 s per scanned page
- No authentication — per case study scope

## For production

- Queue the work and return a job id instead of holding the connection
- Commercial OCR, with the LLM pass standard rather than opt-in
- Golden-file regression suite over a labelled corpus
- Object storage for source documents; Alembic migrations
- Auth, rate limiting, metrics and alerting on the FAIL rate

Deterministic by default · every value traceable to its source row · nothing invented